

AcrossAI · UK AI Energy Intelligence

AI vs Household Appliances

The Complete UK Energy Comparison

How the energy of AI prompts, images and video really compares to kettles, dishwashers, fridges and TVs — with clear UK numbers, methodology and takeaways.

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Why compare AI to appliances?

"An AI prompt uses 0.001 kWh" is a meaningless statement in isolation. The only way to understand AI energy is to compare it with familiar appliances — and once you do, two things become clear at once.

First, an individual AI text prompt is genuinely tiny: far less than boiling a kettle, and often less than an hour of a single LED bulb. Second, that does not make AI energy irrelevant — because the story that matters plays out at data-centre and population scale, where billions of daily queries, model training and cooling add up fast.

The full comparison at a glance

Estimated energy per activity, smallest to largest. AI figures cover typical models; appliance figures are UK household averages.

Activity	Energy (kWh)	Notes
One AI text prompt	~0.001	Typical medium-complexity query
LED bulb (1 hour)	~0.01	10W bulb running for an hour
Charging a smartphone	~0.012	One full charge
One AI image generation	~0.02	Single generated image
Boiling a kettle	~0.1	One full boil (~1.5L)
One AI video generation	~0.2	Short clip, higher end
TV (1 hour, 4K)	~0.2	Large modern screen
Washing machine cycle	~0.8	One standard 40C cycle
Fridge-freezer (1 day)	~1.2	24-hour running average
Dishwasher cycle	~1.2	One standard cycle
Electric oven (1 hour)	~2.0	One hour at ~200C
Tumble dryer cycle	~2.5	One full load

Appliance by appliance

The kettle: the UK's favourite yardstick

Boiling a kettle uses about 0.1 kWh. Almost every UK home owns one, and everyone knows it is a heavy hitter. At roughly 100 text prompts per boil, the kettle instantly reframes AI text use as a small everyday cost rather than something alarming.

Phone charging vs AI images

A full phone charge is about 0.012 kWh. Generating a single AI image (~0.02 kWh) is a little more. Images and video are where AI energy starts to climb — a short AI video (~0.2 kWh) is in the same league as an hour of 4K TV.

The always-on appliances win

A fridge-freezer sips power constantly and totals around 1.2 kWh a day. That single always-on appliance quietly outweighs a heavy day of personal AI text use, which is why 'always-on' beats 'occasional' in the home energy picture.

Laundry: the real household heavyweight

A washing machine cycle (~0.8 kWh) and especially a tumble dryer load (~2.5 kWh) dwarf typical individual AI use. If you want to cut your personal footprint, drying clothes on a line beats worrying about AI prompts by a wide margin.

An hour of TV vs a day of prompts

An hour of large 4K TV uses around 0.2 kWh — comparable to roughly 200 AI text prompts. Entertainment we barely think about routinely outweighs a normal person's daily AI queries.

Even lighting is in the same range

A single LED bulb burning for an hour (~0.01 kWh) uses about as much as 10 typical AI text prompts. Multiply across every bulb in a home and household lighting alone can rival casual AI use.

Text vs image vs video: the AI ladder

Not all AI use is equal. A text prompt (~0.001 kWh) is the floor. An image generation (~0.02 kWh) is roughly 20 times more — about the same as one phone charge. A short video generation (~0.2 kWh) is another order of magnitude, comparable to an hour of 4K TV or two kettle boils.

So the honest headline is: if you only chat with AI, your personal energy use is trivial next to your kettle. If you generate a lot of images and video, it starts to matter — though it is still small compared with laundry, cooking and heating.

Why AI energy still matters

Everything above is about one person. Zoom out and the picture changes. Billions of daily queries across global data centres, the enormous one-off cost of training frontier models, and the electricity and water needed to cool warehouses of GPUs combine into a genuine infrastructure challenge — one that is reshaping the UK grid and data-centre landscape.

The appliance comparison is about personal perspective, not dismissing the wider story. Both can be true: your individual AI use is smaller than your kettle, and AI's aggregate energy demand is a serious national question.

How we calculated these figures

These are illustrative estimates drawn from published research and UK appliance averages, rounded for clarity. Real figures vary with the model, hardware, data-centre efficiency (PUE), tariff and appliance type. AI compute happens in data centres, so it does not appear on your home electricity bill — the comparison shows the underlying energy scale, not a domestic charge.

Run your own numbers

AcrossAI offers free interactive tools to model your own AI energy use and compare it directly with UK appliances and costs:

- AI vs household appliance calculator — tools.acrossai.co.uk/ai-energy-calculators/vs-appliances
- AI query energy calculator — tools.acrossai.co.uk/ai-energy-calculators/query-energy
- AI electricity cost calculator — tools.acrossai.co.uk/energy-cost/ai-electricity-cost-calculator
- AI water consumption calculator — tools.acrossai.co.uk/ai-water-consumption-calculator
- AI energy myth checker — tools.acrossai.co.uk/ai-energy-myth-checker